Lab10: Structural Bioinformatics

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Introduction to the RCSB Protein Data Bank (PDB)

```r
pdb <- read.csv("pdbstats26")
names(pdb)
```

```
[1] "Molecular.Type"   "X.ray"            "EM"               "NMR"
[5] "Integrative"      "Multiple.methods" "Neutron"          "Other"
[9] "Total"
```

```r
head(pdb)
```

```
          Molecular.Type  X.ray    EM   NMR Integrative Multiple.methods
1         Protein (only) 178795 21825 12773         343              226
2 Protein/Oligosaccharide  10363  3564    34           8               11
3             Protein/NA   9106  6335   287          24                7
4     Nucleic acid (only)   3132   221  1566           3               15
```

```
5                    Other     175     25     33            4                 0
6  Oligosaccharide (only)      11      0      6            0                 1
  Neutron Other  Total
1      84    32 214078
2       1     0  13981
3       0     0  15759
4       3     1   4941
5       0     0    237
6       0     4     22
```

```
total_structures <- sum(pdb$Total)
total_structures
```

```
[1] 249018
```

> Q1. What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

X-ray: 80.95% and Electron Microscopy: 12.84%.

```
xray_percent <- sum(pdb$X.ray) / total_structures * 100
em_percent <- sum(pdb$EM) / total_structures * 100

round(xray_percent, 2)
```

```
[1] 80.95
```

```
round(em_percent, 2)
```

```
[1] 12.84
```

> Q2. What proportion of structures in the PDB are protein?

The proportion is 0.979 or about 97.92%.

```
protein_rows <- grepl("Protein", pdb$Molecular.Type)
protein_proportion <- sum(pdb$Total[protein_rows]) / total_structures

round(protein_proportion, 3)
```

```
[1] 0.979
```

```r
round(protein_proportion * 100, 2)
```

```
[1] 97.91
```

> Q3. Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 4,998 HIV-1 protease structures in the current PDB.

Visualizing the HIV-1 Protease Structure

> Q4. Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

We see just one atom per water molecule in this structure because it only shows the oxygen atom of each water molecule. In X-ray crystallography, hydrogen are not usually visible.

> Q5. There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

Yes, I can identify the water molecule. The residue number is HOH 308.

> Q6. Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.

```r
knitr::include_graphics("1HSG-5.png")
```

Introduction to Bio3D in R

```r
library(bio3d)
pdb <- read.pdb("1HSG.pdb")
pdb
```

```
 Call:  read.pdb(file = "1HSG.pdb")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

> Q7. How many amino acid residues are there in this pdb object?

There are 198 amino acid residues.

> Q8. Name one of the two non-protein residues?

MK1

> Q9. How many protein chains are in this structure?

There are 2 protein chains in this structure.

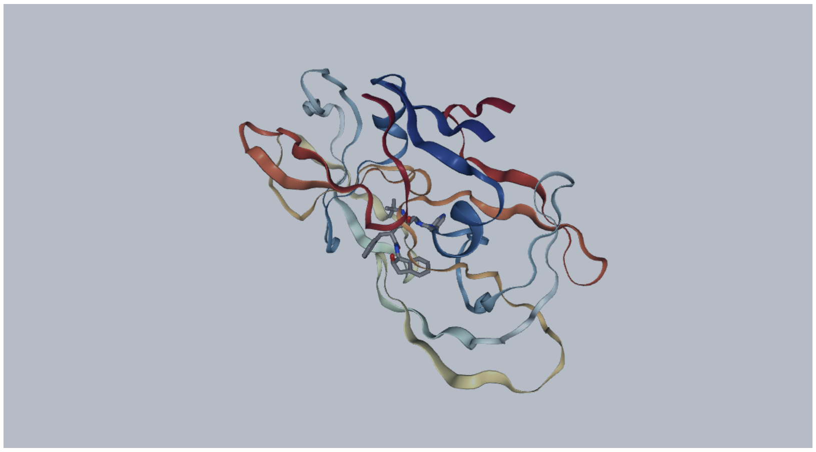
```r
attributes(pdb)
```

```
$names
[1] "atom"   "xyz"    "seqres" "helix"  "sheet"  "calpha" "remark" "call"

$class
[1] "pdb" "sse"
```

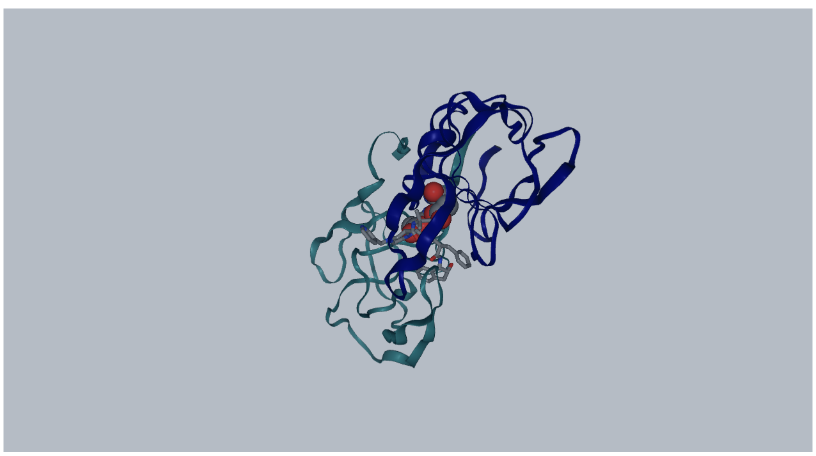
```r
head(pdb$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

```r
# library(bio3dview)
# library(NGLVieweR)

# view.pdb(pdb) |>
# setSpin()

knitr::include_graphics("setspin.png")
```



```r
# sele <- atom.select(pdb, resno=25)
# view.pdb(pdb, cols=c("navy","teal"),
#         highlight = sele,
#         highlight.style = "spacefill") |>
#  setRock()

knitr::include_graphics("rock.png")
```



```r
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

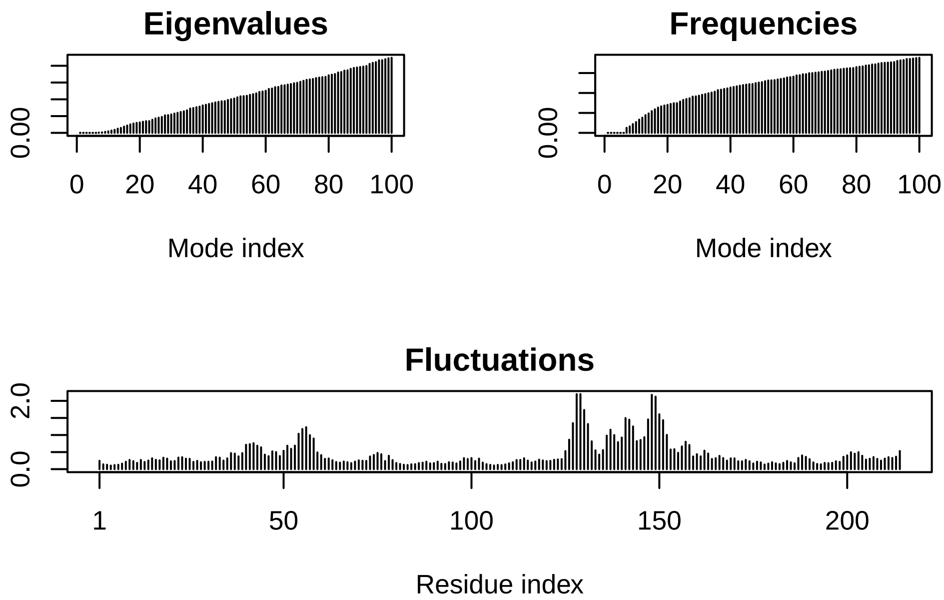
```r
adk
```

```
 Call:  read.pdb(file = "6s36")

   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
m <- nma(adk)
```

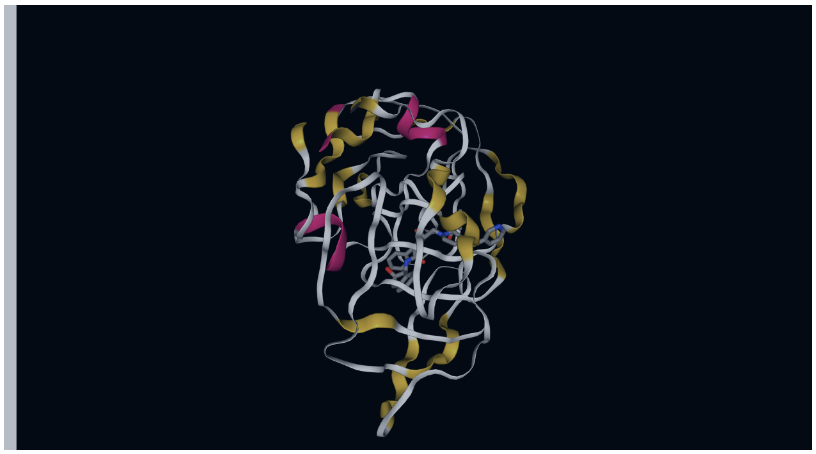
```
 Building Hessian...        Done in 0.009 seconds.
 Diagonalizing Hessian...   Done in 0.193 seconds.
```

```r
plot(m)
```



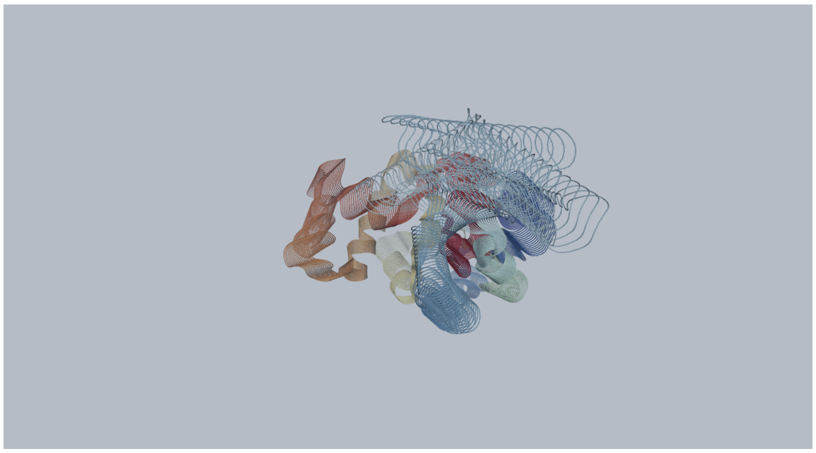
Let's try a custom view

```
# view.pdb(pdb,
#          colorScheme = "sse",
#          backgroundColor = "black")

knitr::include_graphics("view.pdb.png")
```



```
mktrj(m, file="adk_m7.pdb")
```

```
# view.nma(m, pdb=adk)

knitr::include_graphics("nma.png")
```



Comparative Structure Analysis of Adenylate Kinase

> Q10. Which of the packages above is found only on BioConductor and not CRAN?

The msa package is only found in BioConductor.

> Q11. Which of the above packages is not found on BioConductor or CRAN?

The bio3dview is not found.

> Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True

Search and Retrieve ADK Structures

```
library(bio3d)
aa <- get.seq("1ake_A")
```

```
Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta
```

```
Fetching... Please wait. Done.
```

```r
aa
```

```
             1        .         .         .         .         .         60
pdb|1AKE|A   MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
             1        .         .         .         .         .         60

            61        .         .         .         .         .         120
pdb|1AKE|A   DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
            61        .         .         .         .         .         120

           121        .         .         .         .         .         180
pdb|1AKE|A   VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
           121        .         .         .         .         .         180

           181        .         .         .   214
pdb|1AKE|A   YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
           181        .         .         .   214

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

> Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

There are 214 amino acids in this sequence.

```r
hits <- NULL
hits$pdb.id <- c('1AKE_A','6S36_A','6RZE_A','3HPR_A','1E4V_A','5EJE_A','1E4Y_A','3X2S_A','6H
```

```r
# Download releated PDB files
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download
```

```
  |                                                                            
  |                                                                      |   0%
  |                                                                            
  |=====                                                                 |   8%
  |                                                                            
  |===========                                                           |  15%
  |                                                                            
  |================                                                      |  23%
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  |======================================================================| 100%
```

```r
# Align releated PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
```

```
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
....   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...

Extracting sequences

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/6S36_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/6RZE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbs/split_chain/3HPR_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbs/split_chain/5EJE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10   name: pdbs/split_chain/6HAM_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11   name: pdbs/split_chain/4K46_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13   name: pdbs/split_chain/4PZL_A.pdb
```

Annotate Collected PDB Structures

```r
# Vector containing PDB database codes
ids <- basename.pdb(pdbs$id)
```

```r
anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```r
anno
```

```
       structureId chainId macromoleculeType chainLength experimentalTechnique
1AKE_A        1AKE       A           Protein         214                 X-
ray
6S36_A        6S36       A           Protein         214                 X-
ray
6RZE_A        6RZE       A           Protein         214                 X-
ray
3HPR_A        3HPR       A           Protein         214                 X-
ray
1E4V_A        1E4V       A           Protein         214                 X-
ray
5EJE_A        5EJE       A           Protein         214                 X-
ray
1E4Y_A        1E4Y       A           Protein         214                 X-
ray
3X2S_A        3X2S       A           Protein         214                 X-
ray
6HAP_A        6HAP       A           Protein         214                 X-
ray
6HAM_A        6HAM       A           Protein         214                 X-
ray
4K46_A        4K46       A           Protein         214                 X-
ray
3GMT_A        3GMT       A           Protein         230                 X-
ray
4PZL_A        4PZL       A           Protein         242                 X-
ray
       resolution       scopDomain                                      pfam
```

```
1AKE_A       2.00 Adenylate kinase                      Adenylate kinase (ADK)
6S36_A       1.60             <NA>                      Adenylate kinase (ADK)
6RZE_A       1.69             <NA> Adenylate kinase, active site lid (ADK_lid)
3HPR_A       2.00             <NA>                      Adenylate kinase (ADK)
1E4V_A       1.85 Adenylate kinase                      Adenylate kinase (ADK)
5EJE_A       1.90             <NA>                      Adenylate kinase (ADK)
1E4Y_A       1.85 Adenylate kinase                      Adenylate kinase (ADK)
3X2S_A       2.80             <NA>                                        <NA>
6HAP_A       2.70             <NA>                      Adenylate kinase (ADK)
6HAM_A       2.55             <NA>                      Adenylate kinase (ADK)
4K46_A       2.01             <NA>                      Adenylate kinase (ADK)
3GMT_A       2.10             <NA>                                        <NA>
4PZL_A       2.10             <NA>                      Adenylate kinase (ADK)
               ligandId
1AKE_A              AP5
6S36_A MG (2),NA,CL (3)
6RZE_A    CL (2),NA (3)
3HPR_A              AP5
1E4V_A              AP5
5EJE_A           AP5,CO
1E4Y_A              AP5
3X2S_A   JPY (2),AP5,MG
6HAP_A              AP5
6HAM_A              AP5
4K46_A      PO4,AMP,ADP
3GMT_A          SO4 (2)
4PZL_A       GOL,FMT,CA
                                                                        ligandName
1AKE_A                                                    BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
6S36_A                                 MAGNESIUM ION (2),SODIUM ION,CHLORIDE ION (3)
6RZE_A                                              CHLORIDE ION (2),SODIUM ION (3)
3HPR_A                                                    BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
1E4V_A                                                    BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
5EJE_A                              BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A                                                    BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
3X2S_A N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A                                                    BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
6HAM_A                                                    BIS(ADENOSINE)-5'-
```

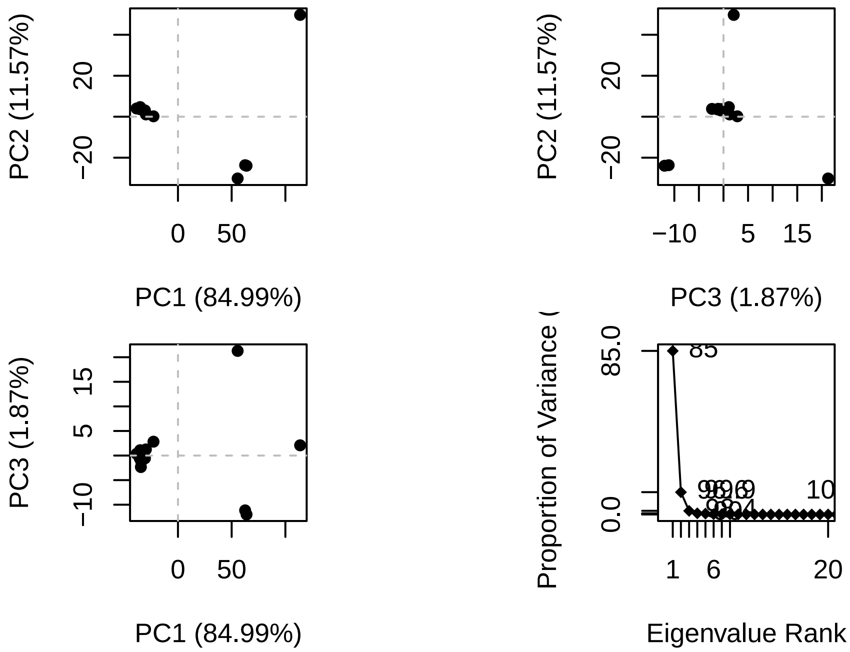
```
PENTAPHOSPHATE
4K46_A                  PHOSPHATE ION,ADENOSINE MONOPHOSPHATE,ADENOSINE-5'-
DIPHOSPHATE
3GMT_A                                                                SULFATE ION (2)
4PZL_A                                                  GLYCEROL,FORMIC ACID,CALCIUM ION
                                                  source
1AKE_A                                 Escherichia coli
6S36_A                                 Escherichia coli
6RZE_A                                 Escherichia coli
3HPR_A                            Escherichia coli K-12
1E4V_A                                 Escherichia coli
5EJE_A           Escherichia coli O139:H28 str. E24377A
1E4Y_A                                 Escherichia coli
3X2S_A         Escherichia coli str. K-12 substr. MDS42
6HAP_A           Escherichia coli O139:H28 str. E24377A
6HAM_A                            Escherichia coli K-12
4K46_A                         Photobacterium profundum
3GMT_A                    Burkholderia pseudomallei 1710b
4PZL_A Francisella tularensis subsp. tularensis SCHU S4

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBI
6S36_A
6RZE_A
3HPR_A
1E4V_A
loop
5EJE_A                                                                                 Crys
1E4Y_A
loop
3X2S_A
conjugated adenylate kinase
6HAP_A
6HAM_A
4K46_A
3GMT_A
4PZL_A                                                                             The crys
                                                     citation rObserved   rFree
1AKE_A             Muller, C.W., et al. J Mol Biology (1992)   0.19600      NA
6S36_A                 Rogne, P., et al. Biochemistry (2019)   0.16320 0.23560
6RZE_A                 Rogne, P., et al. Biochemistry (2019)   0.18650 0.23500
3HPR_A  Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)   0.21000 0.24320
1E4V_A                  Muller, C.W., et al. Proteins (1993)   0.19600      NA
5EJE_A  Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)   0.18890 0.23580
```

```
1E4Y_A                   Muller, C.W., et al. Proteins (1993)   0.17800      NA
3X2S_A                Fujii, A., et al. Bioconjug Chem (2015)   0.20700 0.25600
6HAP_A               Kantaev, R., et al. J Phys Chem B (2018)   0.22630 0.27760
6HAM_A               Kantaev, R., et al. J Phys Chem B (2018)   0.20511 0.24325
4K46_A                    Cho, Y.-J., et al. To be published    0.17000 0.22290
3GMT_A Buchko, G.W., et al. Biochem Biophys Res Commun (2010)   0.23800 0.29500
4PZL_A                       Tan, K., et al. To be published    0.19360 0.23680
         rWork spaceGroup
1AKE_A 0.19600  P 21 2 21
6S36_A 0.15940    C 1 2 1
6RZE_A 0.18190    C 1 2 1
3HPR_A 0.20620  P 21 21 2
1E4V_A 0.19600  P 21 2 21
5EJE_A 0.18630  P 21 2 21
1E4Y_A 0.17800   P 1 21 1
3X2S_A 0.20700 P 21 21 21
6HAP_A 0.22370    I 2 2 2
6HAM_A 0.20311       P 43
4K46_A 0.16730 P 21 21 21
3GMT_A 0.23500   P 1 21 1
4PZL_A 0.19130       P 32
```

Principal Component Analysis

```
# Perform PCA
pc.xray <- pca(pdbs)
plot(pc.xray)
```

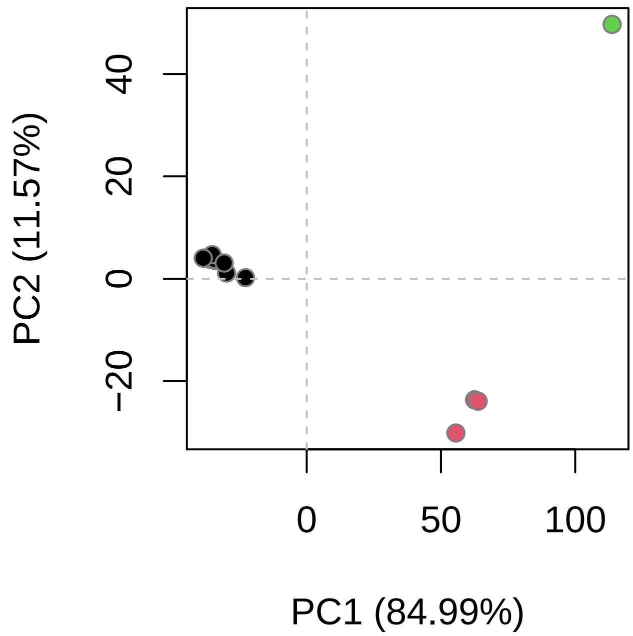



```
# Calculate RMSD
rd <- rmsd(pdbs)
```

```
Warning in rmsd(pdbs): No indices provided, using the 204 non NA positions
```

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



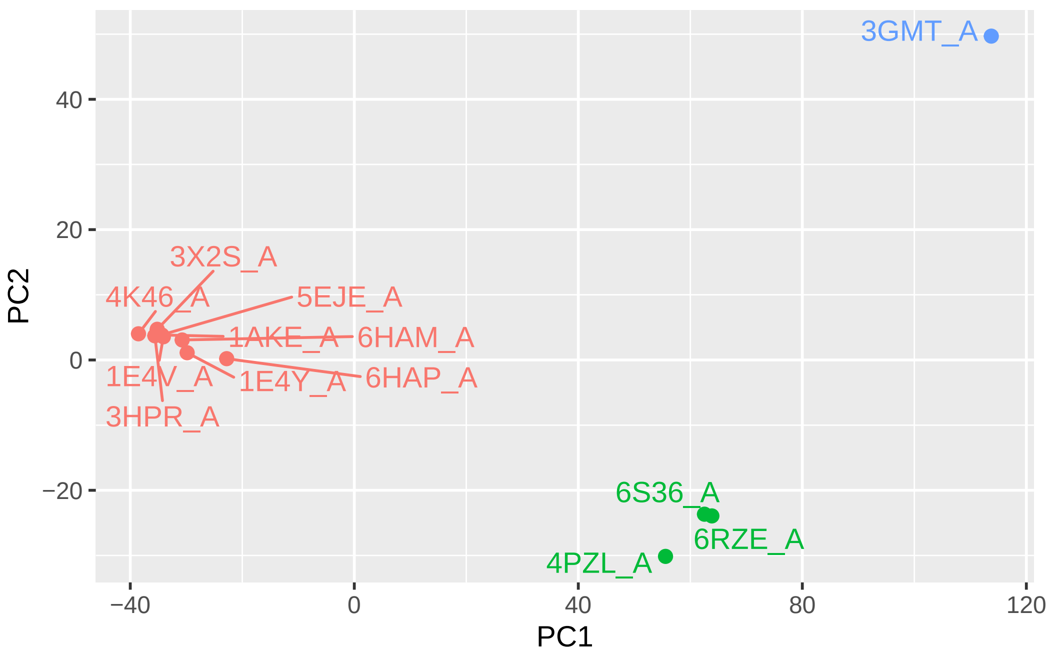
PCA Visualization

```
# Visualize first principal component
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                 PC2=pc.xray$z[,2],
                 col=as.factor(grps.rd),
                 ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```


3GMT_A
3X2S_A
4K46_A
5EJE_A
1AKE_A
6HAM_A
1E4V_A
1E4Y_A
6HAP_A
3HPR_A
6S36_A
6RZE_A
4PZL_A
40
20
0
−20
−40
0
40
80
120
PC1
PC2